

THE FILTERED INTEGRAL SPAN AT WEIGHTS 18 AND 20:

$B(18) = 13$, $B(20) = 14$, AND A SUGGESTED CLOSED FORM

ABSTRACT. Let $\mathcal{M}$ be the algebra of quasimodular forms for $\mathrm{SL}_2(\mathbb{Z})$ with rational Fourier coefficients, $\mathcal{M}_{\mathbb{Z}} = \mathcal{M} \cap \mathbb{Z}[[q]]$, and let $g(\lambda)$ be the Schur MacMahon series. Bachmann and Yu study $B(k) = \min\{n : F_{\leq k}\mathcal{M}_{\mathbb{Z}} \subseteq \sum_{|\lambda| \leq n} \mathbb{Z}g(\lambda)\}$, compute $B(k)$ for $k \leq 16$, and write that these computations suggest the explicit choice $B(k) = \lfloor k/2 \rfloor + \lfloor k/6 \rfloor$. We compute the next two even weights exactly: $B(18) = 13$ and $B(20) = 14$, where that formula gives 12 and 13. The witnesses are two finite chains of lattice indices, each closed by an *attained* index 1; the attained 1 is what makes the two values exact rather than lower bounds. Along the way $[F_{\leq 20}\mathcal{M}_{\mathbb{Z}} : F_{\leq 20}\mathcal{M}_{\mathbb{Z}} \cap \Lambda_{10}] = 5811307335 = 3^{19} \cdot 5$ and the corresponding index at $n = 11$ is $32805 = 3^8 \cdot 5$, while every index we computed at even weights $k \leq 18$ is a power of 3: the 3-primary pattern recorded for $k \leq 16$ does not persist at $k = 20$. The conjecture of Bachmann and Yu that some explicit finite $B(k)$ exists is not at issue; that statement, and the weight-18 and weight-20 instances of their main conjecture $\mathcal{R}_g = \mathcal{M}_{\mathbb{Z}}$, are confirmed here.

1. THE STATEMENT

Let $\mathcal{M}$ denote the algebra of quasimodular forms for $\mathrm{SL}_2(\mathbb{Z})$ with rational Fourier coefficients, $F_{\leq k}\mathcal{M}$ its weight-filtration piece, and

$$\mathcal{M}_{\mathbb{Z}} := \mathcal{M} \cap \mathbb{Z}[[q]], \quad F_{\leq k}\mathcal{M}_{\mathbb{Z}} := F_{\leq k}\mathcal{M} \cap \mathbb{Z}[[q]].$$

With

$$x_m := \frac{q^m}{(1 - q^m)^2} = \sum_{j \geq 1} j q^{mj},$$

the *Schur MacMahon series* is $g(\lambda) := s_{\lambda}(x_1, x_2, \dots)$ for a partition λ ; in particular $A_r := g((1^r)) = e_r(x_1, x_2, \dots)$, and $\mathcal{R}_g := \sum_{\lambda} \mathbb{Z}g(\lambda) = \mathbb{Z}[A_1, A_2, \dots]$ satisfies $\mathbb{Z}[E_2, E_4, E_6] \subset \mathcal{R}_g \subset \mathcal{M}_{\mathbb{Z}}$ [1]. Write

$$\Lambda_n := \sum_{|\lambda| \leq n} \mathbb{Z}g(\lambda), \quad B(k) := \min\{n : F_{\leq k}\mathcal{M}_{\mathbb{Z}} \subseteq \Lambda_n\}.$$

Bachmann and Yu [1] conjecture that there is an explicit function $B(k)$ with $F_{\leq k}\mathcal{M}_{\mathbb{Z}} \subseteq \Lambda_{B(k)}$ for every $k \geq 0$. They then determine the minimal

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$B(k)$ for $k \leq 16$ and write that these computations “suggest the explicit choice”

$$(1) \quad B(k) = \left\lfloor \frac{k}{2} \right\rfloor + \left\lfloor \frac{k}{6} \right\rfloor.$$

The formula is presented with that hedge, as a suggestion drawn from the data, and not as a conjecture in its own right. Their evidence is

k	4	6	8	10	12	14	16
$[F_{\leq k} \mathcal{M}_{\mathbb{Z}} : F_{\leq k} \mathcal{M}_{\mathbb{Z}} \cap \Lambda_{k/2}]$	1	3	3	9	81	729	6561
$B(k)$	2	4	5	6	8	9	10

all seven values of $B(k)$ agreeing with (1); the accompanying text records that “the obstruction to the naive bound $B(k) = k/2$ is 3-primary in every computed case” and that the computation “used 72 coefficients”, which “determine the 67-dimensional space $F_{\leq 20} \mathcal{M}$ ”. Weights 18 and 20 are the next two even weights beyond that computed range.

Our result is that $B(18)$ and $B(20)$ are exactly determined and exceed (1) by 1 each.

Theorem 1. $B(18) = 13$ and $B(20) = 14$, whereas $\lfloor 18/2 \rfloor + \lfloor 18/6 \rfloor = 12$ and $\lfloor 20/2 \rfloor + \lfloor 20/6 \rfloor = 13$. Hence (1) fails at $k = 18$ and at $k = 20$.

Status. What fails is (1), a displayed formula that [1] introduce with the words “suggest the explicit choice” in a paragraph that fences its own evidence at weight 16. It is a hedged suggestion rather than a numbered conjecture, and we do not present it as more than that. The authors’ conjecture that *some* explicit finite $B(k)$ exists is not refuted: Theorem 1 confirms it at $k = 18$ and $k = 20$ with the explicit values 13 and 14, and Corollary 5 below confirms the weight-18 and weight-20 instances of their main conjecture $\mathcal{R}_g = \mathcal{M}_{\mathbb{Z}}$ as well. Nothing here proposes a replacement for (1).

2. THE LATTICES, AND WHY TWO INTEGERS DECIDE THE VALUES

Everything is a lattice computation in a finite-dimensional space, and it is convenient to change basis first.

Lemma 2. *For every $n \geq 0$, the lattice Λ_n is the $\mathbb{Z}$ -span of the Λ -monomials $A_{\mu_1} A_{\mu_2} \cdots A_{\mu_s}$, taken over all partitions μ with $|\mu| \leq n$.*

Proof. Let $\Lambda_{\mathbb{Z}} = \bigoplus_d \Lambda_{\mathbb{Z}}^d$ be the ring of symmetric functions over $\mathbb{Z}$. Fix d . The Schur functions s_{λ} with $|\lambda| = d$ and the elementary monomials $e_{\mu_1} e_{\mu_2} \cdots$ with $|\mu| = d$ are both $\mathbb{Z}$ -bases of $\Lambda_{\mathbb{Z}}^d$ [3], so the transition matrix between them lies in $\mathrm{GL}(\mathbb{Z})$ and the two families have the same $\mathbb{Z}$ -span in each degree, hence the same $\mathbb{Z}$ -span over all degrees $d \leq n$. Applying the ring homomorphism $x_i \mapsto x_i(q)$ carries s_{λ} to $g(\lambda)$ and $e_{\mu_1} e_{\mu_2} \cdots$ to $A_{\mu_1} A_{\mu_2} \cdots$, and carries a $\mathbb{Z}$ -span onto a $\mathbb{Z}$ -span. $\square$

This is what makes the computation feasible: the $\mathbb{Z}$ -span of Lemma 2 needs only $A_1, \dots, A_{14}$ and a list of 373 monomials at $n = 13$ (508 at $n = 14$). We also use, from [1], that $\{g(\lambda) : \lambda_1 \leq 3, |\lambda| \leq n\}$ is a $\mathbb{Q}$ -basis of $F_{\leq 2n}\mathcal{M}$; combined with Lemma 2 over $\mathbb{Q}$ and with $A_r \in F_{\leq 2r}\mathcal{M}$ this gives

$$(2) \quad \begin{aligned} F_{\leq 2n}\mathcal{M} &= \mathbb{Q}\text{-span}\{\text{A-monomials of degree } \leq n\}, \\ \dim_{\mathbb{Q}} F_{\leq 2n}\mathcal{M} &= \#\{(a, b, c) \in \mathbb{Z}_{\geq 0}^3 : a + 2b + 3c \leq n\}, \end{aligned}$$

so $\dim F_{\leq 18}\mathcal{M} = 53$ and $\dim F_{\leq 20}\mathcal{M} = 67$ — the second being the authors' own 67.

For $N \geq 0$ put

$$\begin{aligned} L_N(k) &:= \{f \in F_{\leq k}\mathcal{M} : [q^i]f \in \mathbb{Z} \text{ for } 0 \leq i \leq N\}, \\ i(k, n, N) &:= [L_N(k) : L_N(k) \cap \Lambda_n]. \end{aligned}$$

$L_N(k)$ is a lattice of full rank in $F_{\leq k}\mathcal{M}$ containing $F_{\leq k}\mathcal{M}_{\mathbb{Z}}$ — it is the *dual* lattice of the lattice spanned by the coefficient columns $q^0, \dots, q^N$ of any $\mathbb{Q}$ -basis — and, for $n \geq k/2$, $\Lambda_{k/2} \subseteq L_N(k) \cap \Lambda_n$, so the index is a finite positive integer. For $n < k/2$ the intersection has smaller rank and the index is infinite; correspondingly $B(k) \geq k/2$ always, since $F_{\leq k}\mathcal{M}_{\mathbb{Z}}$ spans $F_{\leq k}\mathcal{M}$ over $\mathbb{Q}$ while $\Lambda_n \otimes \mathbb{Q} = F_{\leq 2n}\mathcal{M}$.

Proposition 3 (the squeeze). *Let $k = 2m$ and $N \geq 0$. For every $n \geq m$,*

$$\Lambda_n \cap F_{\leq k}\mathcal{M} \subseteq F_{\leq k}\mathcal{M}_{\mathbb{Z}} \subseteq L_N(k).$$

If $i(k, n_0, N) = 1$ for some n_0 , then

$$F_{\leq k}\mathcal{M}_{\mathbb{Z}} = L_N(k) = \Lambda_{n_0} \cap F_{\leq k}\mathcal{M},$$

and consequently $i(k, n, N) = [F_{\leq k}\mathcal{M}_{\mathbb{Z}} : F_{\leq k}\mathcal{M}_{\mathbb{Z}} \cap \Lambda_n]$ for every $n \geq m$ and

$$B(k) = \min\{n \geq m : i(k, n, N) = 1\}.$$

Proof. Each A_r lies in $\mathcal{M}_{\mathbb{Z}}$ and $\mathcal{M}_{\mathbb{Z}}$ is a ring, so by Lemma 2 $\Lambda_n \subseteq \mathcal{M}_{\mathbb{Z}}$; intersecting with $F_{\leq k}\mathcal{M}$ gives the first inclusion, and the second is the definition of $L_N(k)$. Now suppose $i(k, n_0, N) = 1$, i.e. $L_N(k) \subseteq \Lambda_{n_0}$. Since $L_N(k) \subseteq F_{\leq k}\mathcal{M}$ we get $L_N(k) \subseteq \Lambda_{n_0} \cap F_{\leq k}\mathcal{M}$, which closes the displayed chain into a triple equality. Then for any n , $L_N(k) \cap \Lambda_n = F_{\leq k}\mathcal{M}_{\mathbb{Z}} \cap \Lambda_n$, so $i(k, n, N)$ is the true index $[F_{\leq k}\mathcal{M}_{\mathbb{Z}} : F_{\leq k}\mathcal{M}_{\mathbb{Z}} \cap \Lambda_n]$; and that index is 1 if and only if $F_{\leq k}\mathcal{M}_{\mathbb{Z}} \subseteq \Lambda_n$, which is the definition of $B(k)$ as a minimum. $\square$

The object exhibited by this note is the pair of chains

	n	9	10	11	12	13	14
(3)	$i(18, n, 300)$	4782969	243	9	3	1	—
	$i(20, n, 360)$	—	5811307335	32805	27	3	1

with $4782969 = 3^{14}$, $243 = 3^5$, $5811307335 = 3^{19} \cdot 5$, $32805 = 3^8 \cdot 5$ and $27 = 3^3$.

Proof of Theorem 1. Only two entries per weight are needed. Take $k = 18$, $N = 300$. Since $i(18, 13, 300) = 1$, Proposition 3 applies with $n_0 = 13$: every entry of the first row of (3) is a true index of $F_{\leq 18}\mathcal{M}_{\mathbb{Z}}$, and $B(18)$ is the least n at which the row equals 1. That is $n = 13$, because $i(18, 12, 300) = 3 > 1$. Hence $B(18) = 13$, while $\lfloor 18/2 \rfloor + \lfloor 18/6 \rfloor = 9 + 3 = 12$.

Take $k = 20$, $N = 360$. Since $i(20, 14, 360) = 1$, Proposition 3 applies with $n_0 = 14$, and $B(20)$ is the least n at which the second row equals 1. That is $n = 14$, because $i(20, 13, 360) = 3 > 1$. Hence $B(20) = 14$, while $\lfloor 20/2 \rfloor + \lfloor 20/6 \rfloor = 10 + 3 = 13$. $\square$

Note where the force of the argument lies. An index > 1 at $n = 12$ on its own would only say $L_N(18) \not\subseteq \Lambda_{12}$, which is weaker than $F_{\leq 18}\mathcal{M}_{\mathbb{Z}} \not\subseteq \Lambda_{12}$ because $L_N(18)$ is a superlattice. It is the *attained* 1 at $n = 13$ that collapses the superlattice onto $F_{\leq 18}\mathcal{M}_{\mathbb{Z}}$ and makes the neighbouring entry an exact index. Correspondingly there is no stabilisation-in- N hypothesis anywhere: N is a computational parameter, not an assumption.

Corollary 4. $[F_{\leq 20}\mathcal{M}_{\mathbb{Z}} : F_{\leq 20}\mathcal{M}_{\mathbb{Z}} \cap \Lambda_{10}] = 5811307335 = 3^{19} \cdot 5$ and $[F_{\leq 20}\mathcal{M}_{\mathbb{Z}} : F_{\leq 20}\mathcal{M}_{\mathbb{Z}} \cap \Lambda_{11}] = 32805 = 3^8 \cdot 5$. Among the indices computed at even weights $k \leq 20$ — those of [1] for $k \leq 16$ and both rows of (3) — these two are the only ones that are not powers of 3; in particular every entry of the $k = 18$ row is a power of 3. So the sentence of [1] that the obstruction is “3-primary in every computed case” is correct as scoped to $k \leq 16$, and the pattern does not persist at $k = 20$. Odd weights are not computed here.

Corollary 5. $F_{\leq 18}\mathcal{M}_{\mathbb{Z}} = \Lambda_{13} \cap F_{\leq 18}\mathcal{M}$ and $F_{\leq 20}\mathcal{M}_{\mathbb{Z}} = \Lambda_{14} \cap F_{\leq 20}\mathcal{M}$. Both right-hand sides lie in $\mathcal{R}_g$, so the weight-18 and weight-20 instances of the Bachmann–Yu main conjecture $\mathcal{R}_g = \mathcal{M}_{\mathbb{Z}}$ hold, and their conjecture that an explicit finite $B(k)$ exists holds at $k = 18$ and $k = 20$ with the explicit values 13 and 14.

Proof. Immediate from the triple equality of Proposition 3 at $n_0 = 13$ and $n_0 = 14$, together with $\Lambda_n \subseteq \mathcal{R}_g$. $\square$

Remark (the computed excess). Write $e(n) := B(2n) - n$. The seven published values give $e = 0, 1, 1, 1, 2, 2, 2$ for $n = 2, \dots, 8$, i.e. $e(n) = \lfloor n/3 \rfloor$, which is (1). Theorem 1 gives $e(9) = e(10) = 4$, so the value 3 is absent from these nine computed values of e , and the step from $B(16) = 10$ to $B(18) = 13$ is +3 where (1) predicts +2. Any replacement guess must fit the nine data points $B(k) = 2, 4, 5, 6, 8, 9, 10, 13, 14$ for $k = 4, 6, \dots, 20$, and the obstruction at $k = 20$ is not 3-primary (Corollary 4).

Remark (truncation dependence at $k = 20$). At $k = 20$ the index $i(20, 10, N)$ computed from the authors’ own 72 coefficients is $87169610025 = 3^{20} \cdot 5^2$, a proper multiple of the value $3^{19} \cdot 5$ obtained at every one of $N = 80, 100, 120, 150, 200, 250, 300$; at $k = 18$ the corresponding index is 3^{14} at all eight of those truncations including $N = 72$. So 72 coefficients do pin the 67-dimensional space $F_{\leq 20}\mathcal{M}$, as [1] say, and do *not* pin its integral lattice.

This is why the argument above is arranged around an attained index rather than around a stable truncation.

3. REPRODUCTION AND VERIFICATION

A referee needs nothing but the definitions above. Fix $k = 2m$ and N . Build $A_r = e_r(x_1, x_2, \dots)$ as integer q -series to order N from $x_m = \sum_{j \geq 1} j q^{mj}$ by the elementary-symmetric recursion; list the Λ -monomials of degree $\leq n$ (Lemma 2); pick $d_m = \dim F_{\leq k} \mathcal{M}$ of the degree- $\leq m$ monomials that are linearly independent, which by (2) is a rank check, and use them as a $\mathbb{Q}$ -basis $F_1, \dots, F_{d_m}$ of $F_{\leq k} \mathcal{M}$. For $c \in \mathbb{Q}^{d_m}$ the q^i coefficient of $\sum_t c_t F_t$ is $\langle c, u_i \rangle$ with $u_i = (F_t[i])_t \in \mathbb{Z}^{d_m}$, so $L_N(k)$ is precisely the dual lattice of $U = \sum_i \mathbb{Z} u_i$: compute a Hermite normal form H of U and read a $\mathbb{Z}$ -basis of $L_N(k)$ off the columns of H^{-1} (each such basis vector must expand to an integral q -series, which is a check). Finally, projecting onto any $\dim F_{\leq 2n} \mathcal{M}$ coefficient positions on which Λ_n injects,

$$i(k, n, N) = \frac{\det \Lambda_n}{\det(\Lambda_n + L_N(k))},$$

a ratio of integer lattice determinants. Every division is exact and no floating-point number occurs anywhere.

What was checked. The program `verify.py` accompanying this note carries out that recipe from scratch in exact integer and rational arithmetic, standard library only, and rechecks every number printed above; its 90 checks all pass. It reproduces the fourteen published integers of [1] for $k \leq 16$, rebuilds $A_1, \dots, A_{14}$ a second time by Newton's identities from the power sums of the x_m , recomputes the four decisive indices in a second, rotated coordinate projection, and runs the eight-truncation sweep quoted in the preceding remark. It covers nothing beyond the claims above: not $k \geq 22$, not odd k , not any replacement closed form, not $\mathcal{R}_g = \mathcal{M}_{\mathbb{Z}}$ outside weights 18 and 20, and not the literature. No explicit element of $F_{\leq 18} \mathcal{M}_{\mathbb{Z}} \setminus \Lambda_{12}$ is exhibited: such an element is determined only modulo Λ_{12} , and the deciding object is the pair of chains (3). The nearest prior work known to us is Craig [2], whose integral generating sets lie in larger algebras of q -multiple zeta values and give no bound on $B(k)$.

REFERENCES

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